

Lecture 2

P1: Convolutional Neural Network.

Convolution $\rightarrow$ Maxpooling $\rightarrow$ Convolution $\rightarrow$ Maxpooling $\rightarrow$ flatten $\rightarrow$ Fully Connected Network
can repeat many times

1. Maxpooling: subsampling resolution $\downarrow$ feature $\downarrow$ 计算量 $\downarrow$

Maxpooling: 取kernel内最大值, 抑制噪声
Avgpooling: 取kernel内平均值

2. Convolution:

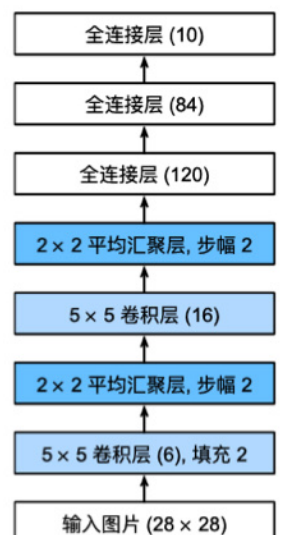
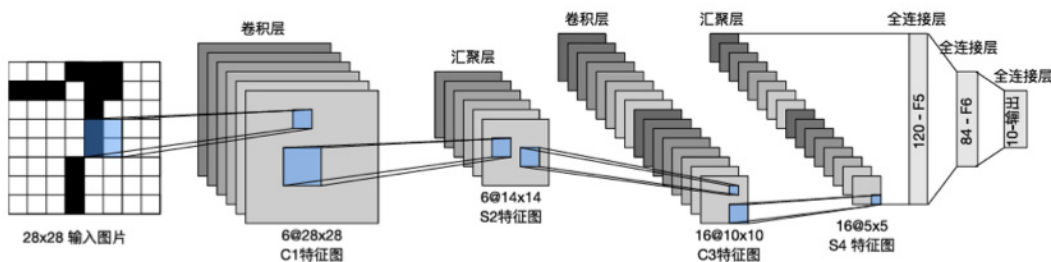
hyper-parameter: Kernel Size, Stride, padding, output channel
width height channel
will not change resolution number of kernel
one channel $\leftrightarrow$ specific pattern {texture, frame, ...}

1x1 Convolutional layer: reduce the number of channel

P2: Modern CNN

1. LeNet

LeNet



全连接层数多 $\rightarrow$ NiN 不可灵活改动 $\rightarrow$ VGG

2. AlexNet: bigger kernel and stride.



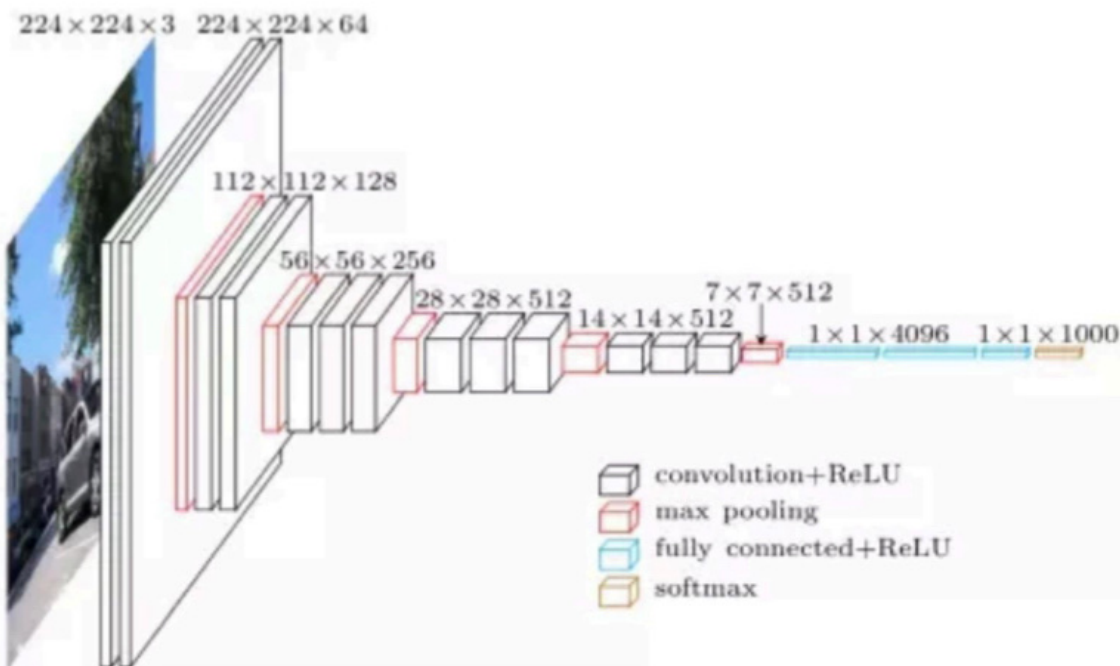
全连接层数多 $\rightarrow$ NiN

循环, 堆叠

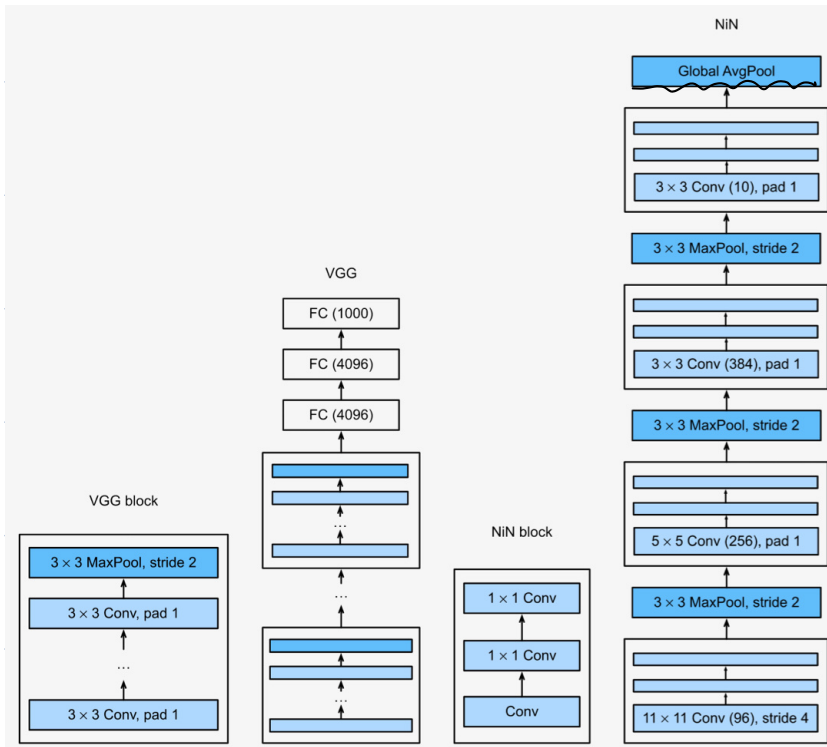
3. VGG: concept of "block"

stacking many blocks, link FCL

e.g. VGG16, 19.



4. NiN : fully connected layer $\rightarrow$ two 1x1 convolutions

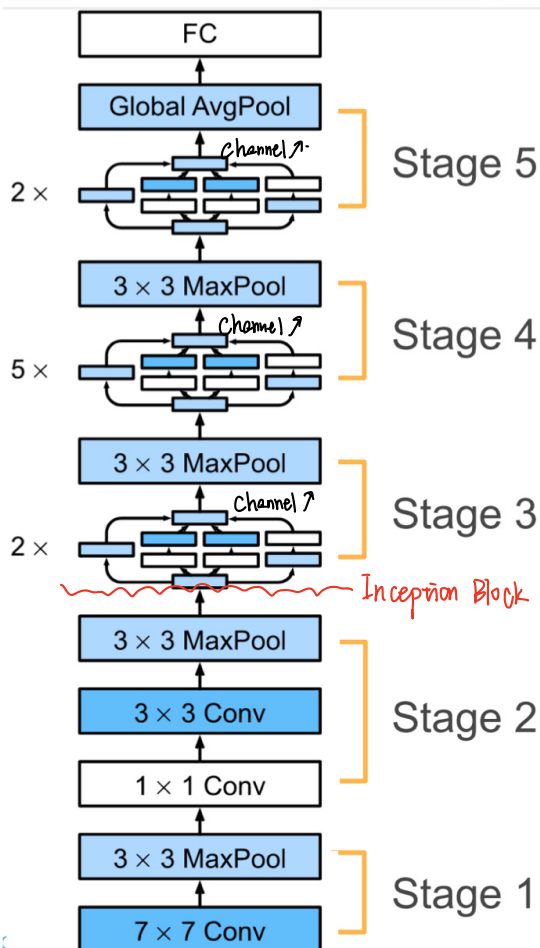


replace fully connected layer
hyperparameter $\downarrow$ overfitting $\downarrow$

交替用 NiN Block 和 Maxpooling

width $\downarrow$ height $\downarrow$ channel $\uparrow$

5. GoogLeNet : import Inception Block



Stage 5

Stage 4

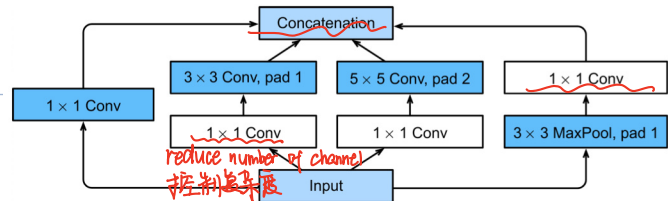
Stage 3

Stage 2

Stage 1

Inception Block

不同层面抽取信息后合并



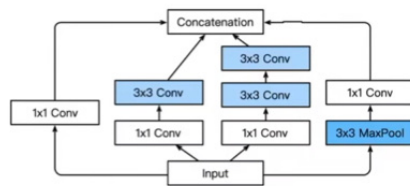
reduce number of channel
控制复杂度

hyperparameter $\downarrow$ 计算复杂度 $\downarrow$

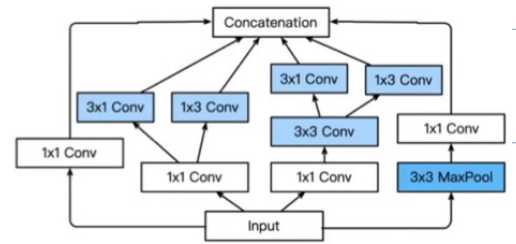
Inception变种: Inception-BN 加 Batch Normalization

Inception-V3 修改 Inception Block, 加残差连接

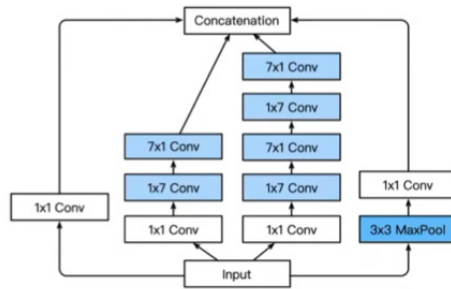
Stage 3



Stage 5



Stage 4



Batch Normalization 批量规范化

后面层训练快, 前面层训练慢.

底层数据一变化, 所有层都要跟着变, 导致后面层要学习多次, 收敛变慢

$$BN(x) = \gamma \cdot \frac{x - \hat{\mu}_B}{\hat{\sigma}_B} + \beta$$

γ : 缩放参数 可学习 β : 平移参数 可学习

$$\hat{\mu}_B = \frac{1}{|B|} \sum_{x \in B} x$$

$$\hat{\sigma}_B^2 = \frac{1}{|B|} \sum_{x \in B} (x - \hat{\mu}_B)^2 + \epsilon$$

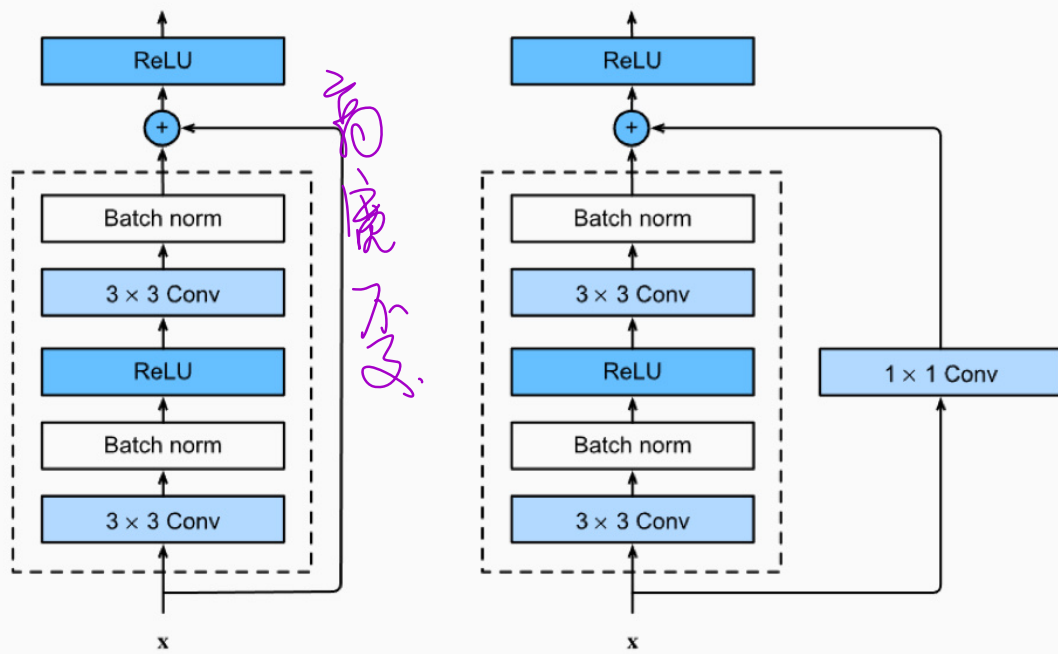
固定小批量里的均值、方差, 加速收敛, 一般不改变模型精度

作用: { 全连接层、卷积层输出上, 激活函数前
全连接层、卷积层输入上

全连接层: 特征值.
卷积层: 通道值.

6. ResNet 残差块加入快速通道 $f(x) = g(x) + x$ 避免梯度消失.

{ 不改变通道数 $\rightarrow f(x) = g(x) + x$
改变通道数 $\rightarrow$ 加 1x1 Conv



高亮减半

Different Residual Blocks



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